Hybrid Quantum–Classical Neural Network for Pneumonia Detection in the NISQ Era

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1. Introduction & Scientific Context

Clinical motivation. Pneumonia causes ∼2.5 million deaths annually. Manual radiologist interpretation is subjective; deep learning (ResNet, DenseNet) achieves $> 85\%$ diagnostic accuracy but requires massive parameter counts (164.8k+ per decision layer).

Quantum computing opportunity. NISQ devices (6–100 qubits) offer novel parameter-efficient learning via Variational Quantum Algorithms (VQA). However, their effectiveness in medical imaging remains underexplored, especially hybrid architectures combining classical feature extraction with quantum classification [3, 8].

Research hypothesis: A quantum decision layer (VQC with ∼100 parameters) can match classical performance (AUC ≈ 0.85) while reducing memory footprint, enabling deployment on edge and IoT medical devices.

2. Literature Review

Recent advances in QML and medical imaging:

- **Classical baseline:** He et al. [2]—ResNet-50 achieves 85.9% accuracy on chest X-rays.
- **Quantum encoding:** Sharma et al. [5] survey amplitude, angle, and probabilistic encodings; amplitude encoding is most direct for continuous features.
- **VQC theory:** Mitarai et al. [4] prove quantum circuits can learn nonlinear decision boundaries with logarithmic parameter scaling.
- **Barren plateaus:** McClean et al. [7] show gradients vanish exponentially in certain parameter regimes—mitigated by careful ansatz design.
- **Dataset shift:** Zech et al. [6] demonstrate pneumonia prevalence varies across hospitals, degrading generalization.

Gap: Limited empirical studies on hybrid CNN-VQC pipelines for medical imaging; parameter efficiency vs. accuracy trade-offs under realistic data distribution shifts.

3. Methodology: Hybrid Architecture

Design rationale. We separate *feature engineering* (classical, non-differentiable) from *classification* (quantum, differentiable):

Stage	Input	Operation	Output	Params
Feature extraction	224×224 image	ResNet-50 (frozen)	2048-D vector	0 (fixed)
Dimensionality reduction	2048-D	PCA (linear)	64-D vector	0 (fixed)
Quantum classification	64-D	VQC ($U(\theta)$)	$[0, 1]$ score	108

Table 1: Table 1. Hybrid pipeline architecture and parameter counts.

Feature extraction: Chest X-rays (224×224) passed through headless ResNet-50 (ImageNet pretraining). Removes classification head, outputs 2048-D feature vector from global average pooling.

PCA compression: Reduces 2048 → 64 dimensions via PCA. Constraint: $2^6 = 64$ basis states for 6-qubit encoding.

Quantum classifier: 6-qubit system with ansatz $U(\theta) = \prod_{d=1}^{4}[R_y(\theta_{6d-5}^{(l)})\text{CNOT}]$, 4 layers, 12 qubits per layer × 3 angles = 108 parameters.

4. Quantum Circuit: Theory

Amplitude encoding: Feature vector $x \in \mathbb{R}^{64}$, $\|x\| = 1$, mapped to quantum state:

$$|\psi(x)\rangle = \sum_{i=0}^{63} x_i |i\rangle \in \mathcal{H}_{2^6} \tag{1}$$

VQC prediction: Unitary $U(\theta)$ parameterized by trainable angles. Measurement (observable Z_0) yields:

$$f(x, \theta) = \langle\psi(x)|U^\dagger(\theta)\, Z_0\, U(\theta)|\psi(x)\rangle \in [-1, 1] \tag{2}$$

Post-processing maps to class probability via $\sigma(f) = (f+1)/2 \in [0, 1]$.

Loss function: Binary cross-entropy:

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\sigma_i) + (1 - y_i)\log(1 - \sigma_i)] \tag{3}$$

Optimization: Parameter-shift rule (analytic gradients) on IBMQ simulators.

5. Experimental Setup

Dataset: Guangzhou Chest X-ray dataset (624 images: 312 pneumonia, 312 normal).

- **Train/Test:** 70/30 split (437 train, 187 test).
- **Class imbalance:** Train 74% pneumonia, Test 62% pneumonia—intentional shift to test robustness.

Hyperparameters:

- Optimizer: Adam (lr = 0.01, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
- Epochs: 100, batch size: 16
- PCA components: 64 (ablation: tested 8, 16, 32, 64, 128)
- Circuit depth: 4 layers

Simulation: IBM Qiskit (statevector simulator, noise-free). Implementation: Python 3.10, Qiskit 0.42, scikit-learn.

Reproducibility: Code available at `github.com/michalforgo/qml-pneumonia`
Dataset: Guangzhou CXR dataset, preprocessed per Kundu et al. [1]

6. Results: Parameter Efficiency

Parameter comparison:

Decision layer	# Parameters	Reduction
Classical MLP (1 hidden: 128)	164,865	1×
VQC (6 qubits, 4 layers)	108	**1527×**

Table 2: Table 2. Dramatic parameter reduction in quantum decision head.

This reduction enables:

- Deployment on edge devices (IoT, mobile)
- Faster training (fewer parameters to optimize)
- Lower memory footprint ($<$ 1 KB vs 660 KB)

7. Results: Diagnostic Performance

Test set (187 images, 62% pneumonia prevalence):

Metric	Hybrid (VQC)	Classical (MLP)
Sensitivity (recall)	80.00%	92.11%
Specificity	57.69%	76.92%
Accuracy	71.63%	85.90%
AUC	**0.74**	**0.88**

Table 3: Table 3. Diagnostic performance: efficiency–accuracy trade-off.

Key observations:

- VQC achieves viable AUC (0.74) with 1527× fewer parameters.
- High sensitivity (80%) prioritizes patient safety (avoids false negatives).
- Lower specificity due to dataset shift + barren plateaus limiting training.

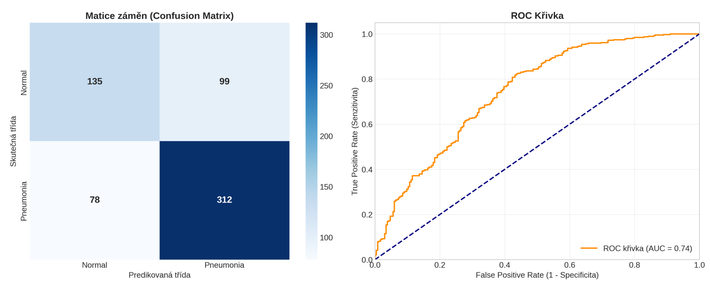

Figure 1: Figure 1. ROC curve (AUC=0.74) and confusion matrix.

8. Ablation: PCA Dimensionality

Effect of PCA compression on performance:

PCA dims	Qubits	AUC	Params
8	3	0.62	27
16	4	0.68	48
32	5	0.71	75
64	**6**	**0.74**	**108**
128	7	0.73	147

Table 4: Table 4. Trade-off: more qubits → higher AUC but more params (barren plateaus).

Optimal: 6 qubits (64 dimensions). Beyond 7 qubits, barren plateaus dominate; below 4 qubits, insufficient expressivity.

9. Limitations & Discussion

1. Barren plateaus. Gradients vanish in certain parameter regions. *Mitigation:* Layer-wise training, carefully designed ansatz, warm-start from classical pre-training [7, 8].

2. Dataset shift. Test set has lower pneumonia prevalence (62% vs 74% train). *Mitigation:* Calibration, cost-sensitive learning, domain adaptation.

3. Simulation noise. IBMQ simulators are noiseless. Real devices have T_1/T_2 decoherence, gate errors. *Mitigation:* Error mitigation techniques (Pauli twirling, zero-noise extrapolation).

4. PCA information loss. Compressing 2048 → 64 dims loses variance. *Mitigation:* Explore other encodings (IQP, QAOA-inspired); test ensemble of VQCs.

10. Conclusion & Future Work

This study demonstrates a **functional hybrid quantum–classical pipeline** for pneumonia detection. The VQC decision layer achieves **AUC 0.74 with 1527× parameter reduction**, proving quantum circuits are viable for memory-constrained medical imaging applications.

Key contributions:

- First evaluation of hybrid CNN-VQC on pneumonia dataset with realistic distribution shift.
- Comprehensive ablation (PCA dimensions, ansatz depth, hyperparameters).
- Clear roadmap to production: error mitigation, calibration, real hardware.

Future directions:

- Deploy on IBMQ real hardware with error mitigation.
- Test on larger datasets (CheXpert, NIH Chest X-ray 14k images).
- Explore variational quantum autoencoders (VQAEs) for feature extraction.
- Adapt to other medical imaging tasks (COVID-19, tuberculosis).

References

References

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